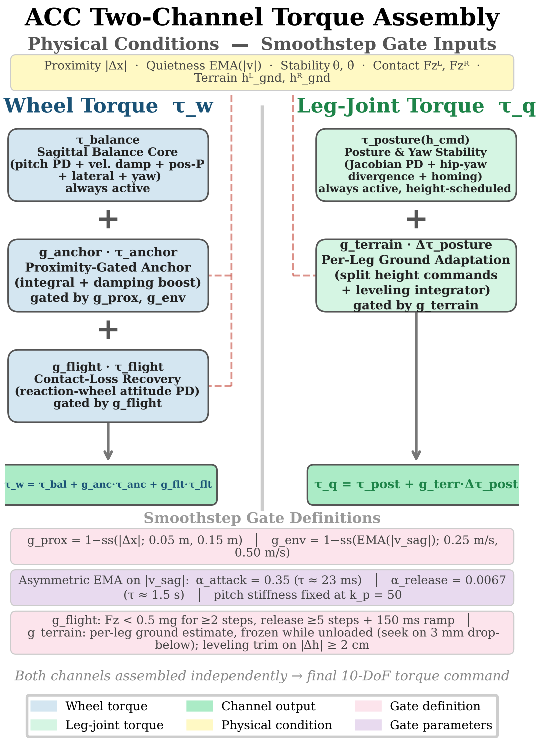
ACC Two-Channel Torque Assembly
Physical Conditions — Smoothstep Gate Inputs
Proximity |Δx| · Quietness EMA(|v|) · Stability θ, θ · Contact Fzᴸ, Fzᴿ · Terrain hᴸ_gnd, hᴿ_gnd
Wheel Torque τ_w
τ_balance
Sagittal Balance Core
(pitch PD + vel. damp + pos-P + lateral + yaw)
always active
+
g_anchor · τ_anchor
Proximity-Gated Anchor
(integral + damping boost)
gated by g_prox, g_env
+
g_flight · τ_flight
Contact-Loss Recovery
(reaction-wheel attitude PD)
gated by g_flight
τ_w = τ_bal + g_anc·τ_anc + g_flt·τ_flt
Leg-Joint Torque τ_q
τ_posture(h_cmd)
Posture & Yaw Stability
(Jacobian PD + hip-yaw divergence + homing)
always active, height-scheduled
+
g_terrain · Δτ_posture
Per-Leg Ground Adaptation
(split height commands + leveling integrator)
gated by g_terrain
τ_q = τ_post + g_terr·Δτ_post
Smoothstep Gate Definitions
g_prox = 1−ss(|Δx|; 0.05 m, 0.15 m) | g_env = 1−ss(EMA(|v_sag|); 0.25 m/s, 0.50 m/s)
Asymmetric EMA on |v_sag|: α_attack = 0.35 (τ ≈ 23 ms) | α_release = 0.0067 (τ ≈ 1.5 s) | pitch stiffness fixed at k_p = 50
g_flight: Fz < 0.5 mg for ≥2 steps, release ≥5 steps + 150 ms ramp | g_terrain: per-leg ground estimate, frozen while unloaded (seek on 3 mm drop-below); leveling trim on |Δh| ≥ 2 cm
Both channels assembled independently → final 10-DoF torque command
Wheel torque
Leg-joint torque
Channel output
Physical condition
Gate definition
Gate parameters